

EARTHPRESS · DESIGN DECISION RECORD

EarthPress TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	EPR-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/decisions/0001-trl2-review-decisions.md @ 0dc0dd4		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, the items that remain open, and new items from EPR-CAL-001

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. Items 1 to 7 and 9 are adopted as recommended for TRL 3 work, pending Amish's review. Item 8 and new items 10 to 16 remain proposed, awaiting Amish.

■ Context

The TRL 2 review note (docs/REVIEW.md , session of 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", eight of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the EarthPress items one by one. Under that instruction, each item that has a recommendation is adopted as recommended so that the TRL 3 work can proceed, and stays open for his review. Items without a recommendation stay open. TRL 4 is on hold by Amish's instruction.

EarthPress uses none of the shared components in this batch (FieldNode, CellGuard, MotionCore, ThermaCart, TwinKit, CalRig) or SwapCell, so there are no cross-repo interfaces to keep consistent.

■ Options considered

The options for items 1 to 9 are in docs/REVIEW.md (TRL 2 section). Items 10 to 16 are new at TRL 3 and come from EPR-CAL-001; their options are in Table 2.

■ Decision

Table 1. Items adopted for TRL 3.

#	Item	Status	Where it now lives
1	Mechanism: bottom piston with a toggle and one lever (CINVA-Ram pattern); the bottle-jack variant documented later	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review	EPR-PRC-001 v0.3, <code>cad/src/model.py</code>
2	Lever layout: raise the pivot for a longer arc and a higher grip at the end of the stroke	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. EPR-CAL-001 section 3 shows the specific figures (800 mm pivot, 100° arc) do not work with a 1.65 m grip radius, because the grip would pass below the ground; the intent is met with a low pivot and a 59° arc from a nearly upright start (grip 1.89 to 0.86 m). The geometry change is item 16a	EPR-CAL-001 section 3, <code>cad/src/model.py</code>
3	Block: flat 290 x 140 x 90 mm (Auroville module); interlocking inserts a later option	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review	EPR-REQ-001 v0.3, EPR-PRC-001 v0.3
4	Compaction target: 2 MPa	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review	EPR-REQ-001 R2
5	Stabilizer: 5 % cement by dry mass, lime for clay-rich soils, confirmed with partner soils	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review	EPR-PRC-001 v0.3, EPR-REQ-001 R10
6	Meeting R7: 12 mm base plate and a bolted mold (rather than relaxing R7)	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. The frame is now two bolted pieces (36.6 and 27.7 kg) and no piece exceeds 37 kg, but the total is 187 kg against 140 kg (item 13)	<code>cad/src/model.py</code> , EPR-CAL-001 section 8
7	Soil test kit: field tests only at TRL 3	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review	EPR-PRC-001 v0.3, BOM item 9
9	Budget and wording: no change to <code>budget_usd</code> (\$450); pitch and problem unchanged	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. <code>budget_usd</code> stays at \$450; the BOM is now over it (item 14)	<code>project.yaml</code> (unchanged)

No pitch or problem rewording was recommended at TRL 2, so `project.yaml` and `README.md` keep the TRL 2 wording.

Items that remain open

Table 2. Open items, all Proposed, awaiting Amish.

#	Item	Options and recommendation	Status
8	First co-design partner and region	Examples at TRL 2: an earth-building NGO in East Africa or India. No recommendation was made	Proposed, awaiting Amish
10	Ejection	The TRL 2 concept ejected by pulling the lever past the compaction point; a toggle ending near straight cannot lift the piston another 90 mm. Options: (a) a seesaw eject lever on the +X side, driven by the same removable lever in a second socket, with a 200 mm lost-motion slot in the push rod (in the model); (b) a bell crank that ejects on an upward lever stroke (awkward to push up at 1.9 m and above); (c) a permanently fitted second lever. Recommendation: (a)	Proposed, awaiting Amish; (a) is in the model
11	Two operators for R2	Peak pull is 668 N; one operator at 500 N stalls at 0.53 MPa. Options: (a) two people on a T-handle (334 N each; in the model); (b) accept one operator pulling about 670 N with body weight; (c) lower the target pressure. Recommendation: (a)	Proposed, awaiting Amish
12	R5 wording	The 0.9 to 1.6 m band allows only 24.5° of arc. Recommendation: reword R5 as 500 N or less per operator, grip between 0.8 and 1.9 m, with the peak force at 1.0 m or higher (the design gives 1.16 m)	Proposed, awaiting Amish; R5 unchanged in EPR-REQ-001
13	R7 total mass	187 kg against 140 kg. Options: (a) relax the total to 190 kg and keep the 50 kg piece limit; (b) lighten (flat-bar columns, 16 mm lid plate, 10 mm mold walls), perhaps 15 kg, then re-check; (c) both. Recommendation: (a), since the piece limit is what governs moving it	Proposed, awaiting Amish; R7 unchanged
14	Budget	BOM \$488 against \$450. Options: (a) raise <code>budget_usd</code> to \$500; (b) cost down: plain pins without bushes (about \$30 less, but fails R8), local offcut steel, lighter frame; (c) hold \$450 until real quotes. Recommendation: (a). <code>budget_usd</code> is not changed; the recommended figure is \$500, awaiting Amish	Proposed, awaiting Amish
15	Fill control	The fill must be 0 to +2.7 % of nominal (0 to 206 g) for 2 MPa at the stop. Proposal: weigh every fill on a 10 kg scale with 50 g divisions (in the BOM) to about +1.3 % ± 100 g, and put the target mass on the test chart	Engineering proposal, awaiting Amish
16	Other TRL 3 engineering proposals	(a) Lever pivot at $x = -468$, $z = 266$ mm, 1.7 m lever of 2 in schedule 40 pipe, mold rim at 940 mm; (b) crank stop at the 6° end angle, which limits the linkage to 173 kN (the structural design case); (c) ribbed lid (the TRL 2 22 mm plate reaches 275 MPa at 2 MPa); (d) end pawl against 47 J of kickback; (e) guard over the crank, connecting link and knee; (f) case-hardened steel bushes; (g) R4 at 296 blocks a day, 1 % short, to be revisited when cycles are timed	Engineering proposals, awaiting Amish

■ Consequences

- EPR-PRB-001, EPR-PRC-001 and EPR-REQ-001 move to v0.3. The design choices in Table 1 are no longer described as proposals, but each stays open for Amish's review.
- `budget_usd` stays at \$450; R13 is not met against it (\$488) and would be met against the recommended \$500.
- R5, R7 and R13 are not met on paper; R2 and R4 are at risk; R3, R8, R10 and R14 cannot be verified before block and wear tests, which are TRL 4 work and on hold.